6 b output.sci (C:\Users\LENOVO\Documents\6 b output.sci) - SciNotes

*6 b output.sci

```
// Test Case: Analysis of a Discrete-Time Linear System

// Define parameters
N = 20; // Number of samples
n = 0:N-1; // Discrete time index vector
x_n = ones(1, N); // Input sequence: Unit step sequence
y_n = zeros(1, N); // Output sequence initialization
// Define the difference equation: y[n] = 0.5y[n-1] + x[n]
for k = 2:N
y_n(k) = 0.5 * y_n(k-1) + x_n(k);
end
// Plot the input sequence
clf; // Clear the current figure
subplot(2,1,1);
plot2d (n, x_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Input Sequence: Unit Step Sequence');
// Plot the output sequence
subplot(2,1,2);
plot2d(n, y_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Output Sequence: Response of the Discrete-Time System');

```

